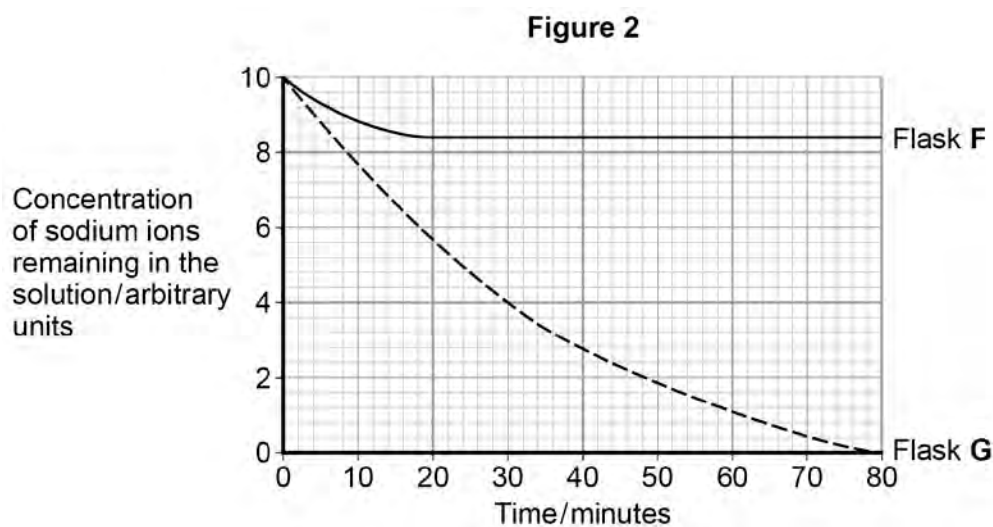


2

A scientist investigated the uptake of sodium ions by animal tissue. To do this, he:

- used two flasks, **F** and **G**
- put equal masses of animal tissue into each flask
- added equal volumes of a solution containing sodium ions to each flask
- added to flask **F** a solution of a substance that prevents the formation of ATP by cells
- measured the concentration of sodium ions **remaining** in the solution in each flask.

Figure 2 shows his results.



0 2

. 1

Calculate the rate of uptake of sodium ions by the tissue in flask **G** during the first 20 minutes of this investigation.

[1 mark]

Answer = _____ arbitrary units per minute

0 2 . **2** The scientist concluded that the cells in flask **G** took up sodium ions by active transport. Explain how the information given supports this conclusion.

[4 marks]

[Extra space]

0 2 . **3** The curve for flask **F** levelled off after 20 minutes. Explain why.

[2 marks]
